

Learning When and How to Update Memory for Video Object Segmentation

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Abstract—Recent progress in semi-supervised video object segmentation has largely hinged on memory-based methods. However, when faced with increasingly tough challenges emerging in complex scenarios, such as fundamental semantic transformations and severe spatial deformations, the fixed-interval memory update mechanism usually adopted in these memory-based methods is insufficient to align with the pivotal moments of object changes. This inflexible mechanism motivates us to design an adaptive memory update mechanism in response to the semantic-spatial changes of target objects. To this end, we propose a novel Change-Sensitive Network (CSNet) to learn when and how to update memory to effectively address intricate challenges in complex scenarios. Specifically, we first design an Adaptive Perception-Capture module with a hierarchical contrastive learning loss to determine when to update memory moments by measuring the extent of object changes, thus dividing entire videos into different object-change clips. To further extract and highlight object changes to assist in the segmentation of frames after changes occur, we construct Dynamic Memory Update modules to redefine how to update memory by smoothly retaining the object prototypes within clips and dynamically amplifying the object variations across clips. Extensive experiments demonstrate that our proposed CSNet exhibits clear superiority when evaluated on eight datasets covering three kinds: common, complex and long-video datasets.

Index Terms—Video object segmentation, contrastive learning, memory update mechanism.

I. INTRODUCTION

THE field of semi-supervised video object segmentation (Semi-VOS) aims to segment objects in a video sequence using the initial frame's object mask as a reference. Early attempts for Semi-VOS are mainly based-on online-learning [1], [2], [3], [4], [5], mask propagation [6], [7], [8], [9] and matching [10], [11], [12], [13], [14]. Recently, memory-based methods [15], [16], [17], [18], [19], [20], [21], [22], [23], [24],

[25] have gained prominence in addressing this task. These memory-based methods leverage historical frames to enhance the segmentation of current frames, leading to significant improvements on common datasets such as DAVIS [26] and YouTube-VOS [27].

However, existing Semi-VOS methods [15], [16], [18], [20], [22], [24], [25], [28] using memory networks typically memorize appearance cues at fixed interval, for example, updating memory every five frames. These methods tend to determine the update intervals according to the performance, and such fixed-interval memory update mechanism as shown in Fig. 1(a) becomes fragile when faced with complex scenarios, leading to a significant decrease in accuracy when evaluated on emerging complex datasets including VOST [29] and MOSE [30].

To effectively deal with complex scenarios, we further observe the characteristics of videos in emerging complex datasets [29], [30]. As illustrated in Fig. 1, these videos exhibit two primary challenges: **fundamental semantic transformations**: never-appeared meat is taken out of the plastic bag from t_1 to t_3 moments, and **severe spatial deformations**: the plastic bag is obscured by the hands significantly from t_4 to t_6 moments. When objects start to change significantly, memorizing changed features at t_2 and t_5 moments as shown in Fig. 1(b) is of great significance for segmenting following frames undergone current object changes. Therefore, the current fixed-interval memory update mechanism cannot sufficiently align with the pivotal moments of object changes, highlighting a demand for determining the memory update moments based on the extent of object semantic-spatial changes.

Meanwhile, previous memory-based methods [17], [19], [21], [23], [31], [32] usually update memory by extracting object prototypes from the received update frames to prevent the memory from expanding infinitely. For example, XMem [17] update its long-term memory through selecting representative prototypes based on the frequency of usage. However, this form of memory update lacks the emphasis on object variations before and after changes. As there are various semantic-spatial changes occurred in complex scenarios, it is crucial to highlight object variations during memory update. Therefore, compared to passively extracting prototypes, proactively acquiring object variations across different clips of videos can be a breakthrough solution to address intricate object changes.

With the above considerations in mind, we propose a Change-Sensitive Network (CSNet) with an adaptive memory update mechanism to learn when and how to update memory. Firstly, we

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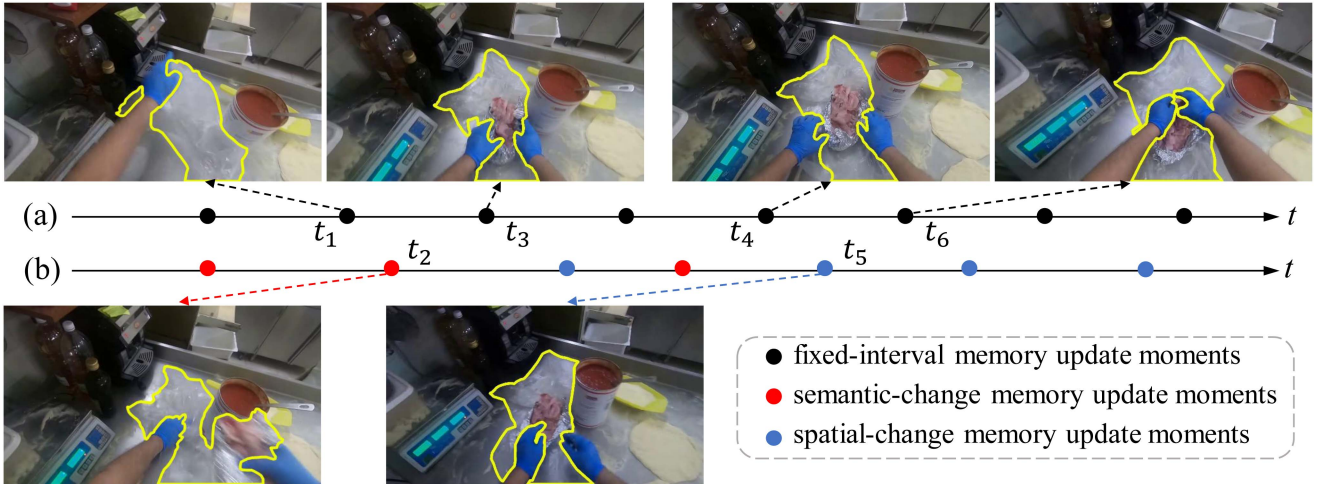


Fig. 1. Comparison of (a) fixed-interval memory update mechanism and (b) our proposed adaptive memory update mechanism. When encountering fundamental semantic transformation (the never-appeared meat is taken out of the plastic bag from t_1 to t_3 moments) and severe spatial deformation (the plastic bag and the meat are obscured by the hands significantly from t_4 to t_6 moments), our proposed mechanism can perceive and capture the moments of semantic and spatial change at t_2 and t_4 moments respectively, then adaptively update memory to address these dramatic changes as soon as they begin to deal with complex scenarios. The yellow area in the frames indicates the target objects.

design an Adaptive Perception-Capture (APC) module to learn when to update memory by discriminatively distinguishing the extent of semantic transformations and spatial deformations of target objects. To pay more attention to semantic-spatial changes of target objects and mitigate the impact of background changes, we design a hierarchical contrastive learning loss at frame, object and pixel levels for our APC module to generate effective representations by highlighting as much semantic and spatial informations of target objects as possible. Through keenly perceiving and capturing the extent of object semantic-spatial changes in our APC module, the memory update moments can be determined and the entire videos can be divided into different object-change clips.

To further adapt memory according to object changes, we construct Dynamic Memory Update modules to learn how to update memory based on both intra-clip and inter-clip object changes. When object changes are insignificant decided by our APC module and maintain in the same clip, a Short-term Memory Modulation module is constructed to modulate object prototypes within clips based on their similarities and extract pivotal informations to be stored in the short-term memory. When the semantic or spatial changes of objects are so prominent that our APC module determines to enter into a new clip, a Long-term Memory Augmentation module is constructed to perceive object variations across clips and enhance crucial differences before being stored in the long-term memory.

Based on our proposed Adaptive Perception-Capture module and Dynamic Memory Update modules, our proposed CSNet can adaptively identify the key moments of significant object changes at semantic and spatial aspects, and dynamically adapt short-term and long-term memory according to various object changes. To validate the effectiveness of our proposed CSNet, extensive and comprehensive experiments are conducted on eight Semi-VOS datasets including four common datasets [26], [27], [33], two complex datasets [29], [30] and one long-term

datasets [34], [35]. The experimental results show that our proposed CSNet can achieve remarkable performances on these eight datasets.

In general, the main contributions of our work are summarized as follows:

- We propose a novel Change-Sensitive Network (CSNet) for actively learning when and how to update memory to effectively address emerging challenges in complex scenarios.
- We design an Adaptive Perception-Capture module with a hierarchical contrastive learning loss to master the nuances of object semantic-spatial changes and discern temporal patterns of various kinds of videos.
- We construct Dynamic Memory Update modules including a Short-term Memory Modulation module to extract pivotal prototypes within clips and a Long-term Memory Augmentation module to enhance object variances across clips, which effectively adapt memory based on the extent of object changes.
- Extensive experiments clearly show that our proposed CSNet can achieve significant improvements compared to existing state-of-the-art methods on common, complex and long-term datasets.

II. RELATED WORK

A. Semi-Supervised Video Object Segmentation

Video Object Segmentation [1], [13], [15], [16], [17], [20], [36], [37], [38], [39], [40], [41], [42], [43] has developed for many years. Different from unsupervised video object segmentation that utilizes spectral clustering [44], [45] or long-short temporal attention [46] to capture space-time dependencies, which can segment target objects without initial specifying, semi-supervised video object segmentation (Semi-VOS) can utilize the initial frame's object mask as a reference during

segmentation. There are early attempts for Semi-VOS including online-learning [1], [2], [3], [4], [5], [18], [20], [47], [48], propagation-based [6], [7], [8], [9] and matching [10], [11], [12], [13], [14], [49], [50], [51], [52], [53], [54] methods. **Online-learning methods** [1], [2], [3], [4], [5], [47], [48] segment the target objects through online fine-tuning the given annotated objects. For example, OSVOS [1] fine-tunes its pre-trained network with first given frame and mask to segment subsequent frames. **Propagation-based methods** [6], [7], [8], [9], [55], [56] refine current segmentation masks through propagating preceding masks. For example, MHP [56] constructs propagation tree for confirming spatial location of objects in current frames. **Matching-based methods** [10], [11], [12], [13], [14], [49], [50], [51], [52], [53], [54], [57], [58] performs pixel-level similarity calculation and pairing to classify label of pixels. For example, RANet [10] computes and selects similarity maps between current and historical features with correlation and ranking attention module.

With the emergence of the instructive STM [15] model, **memory-based methods** [15], [17], [23], [24], [31], [32], [59], [60] begin to greatly improve the accuracy when evaluated on the relatively simple common datasets [26], [27], [33]. As historical frames can act as strong clues to guide the segmentation of the current frame, STM [15] continuously updates the target-specific features in the memory network and utilizes non-local matching between query and memory features to make full use of historical features to effectively improve the performance. EGMN [57] designs a graph memory network which stores frames as nodes and capture cross-frame correlations by edges, and proposes a learnable controllers to ease memory reading and writing. Moreover, STCN [16] reuses target-agnostic features to calculate similarity and improves the calculation method from dot product to negative squared Euclidean. IMSFR [61] designs an isogenous memory sampling and STCM [62] proposes sparse group memory reader to improve efficiency. More sparse memory networks [63], [64], [65] are further proposed to reduce video-level temporal redundancy. Meanwhile, AOT [20], DeAOT [18] and AOST [66] replace affinity calculation with short and long-term attention and construct an identification mechanism to simultaneously process multiple objects matching and segmentation efficiently. Furthermore, ISVOS [67] and Cutie [19] pay more attention to the object-related features and design different architectures to enhance object prototypes.

However, these memory-based methods are still too weak to confront emerging challenges in the complex datasets [29], [30], [34]. As the target objects in these complex datasets with intricate scenarios can undergo fundamental semantic transformations and severe spatial deformations that are significantly different from the previous common datasets, only relying on appearance clues makes these existing memory-based methods fragile due to the missing of key informations during object changes, leading to a sharp decline in overall performance when evaluated on these complex datasets.

B. Memory Update Mechanism

Memory-based Semi-VOS methods [17], [21], [23], [24], [25], [31], [32], [59], [61], [62], [63], [64], [65], [68], [69],

[70], [71] have designed lots of memory update mechanisms to manage memory. **Memory pruning methods** [23], [24], [32] reduce redundancy of features stored in memory. For example, compared to retaining all the features of memory frames in STM [15], AFB-URR [32] fuses features with higher similarity to existing features and only retains less similar features to reduce redundancy, which can update memory bank by dynamically absorbing new features and discarding obsolete features. Based on successfully reusing the target-agnostic key in the memory bank in STCN [16], SWEM [24] further reduces the redundancy of both inter-frame and intra-frame features with weighted expectation-maximization.

Multi-stage memory update methods [17], [19], [34] are further designed to memorize object features more comprehensively across different temporal-stages. For example, XMem [17] delicately designs multi-stage memory to segment the current frame, which has almost maximized the role of memory in different stages. In multi-stage memory update mechanism, XMem consolidate long-term memory by sampling representative key-value feature pairs from working memory, and recurrently update the sensory memory with GRU [72]. However, these memory update mechanisms passively update memory according to the fixed-interval and rely too much on the historical features throughout the entire videos, which makes these memory-based methods fail to extract object prototypes to be stored in memory during various object changes.

When faced with the challenges in complex datasets, it is not enough for existing memory-based Semi-VOS methods to only draw on historical appearance clues. In this work, we rethink the cores of memory-based methods: when and how to update memory. Our proposed adaptive memory update mechanism in our Change-Sensitive Network can continuously follow the semantic and spatial changes of target objects, dynamically update the memory to better adapt to the object changes and effectively address the challenges of emerging complex scenarios.

III. APPROACH

In this section, we first revisit existing memory-based methods from the perspective of memory update mechanism. Then we present our proposed Change-Sensitive Network (CSNet) by introducing its overview, and core modules including an Adaptive Perception-Capture (APC) module and Dynamic Memory Update modules, which are designed to address when and how to update memory respectively.

A. Revisit Memory-Based Methods

Memory-based methods use historical clues stored in the memory to guide the segmentation of the current frame. Formally, given a video with T frames $\mathcal{I} = \{I_t\}_{t=1}^T$, the current frame I_t is encoded and embedded to generate the query embedding K_t^q . The memory key and value embeddings K^m, V^m are retrieved from the stored memory $M_{t-\delta}$. Then the query embedding is transferred from object-agnostic key K_t^q to object-specific value V_t^q via K^m, V^m as:

$$V_t^q = V^m \text{Softmax}(\langle K_t^q, K^m \rangle), \quad (1)$$

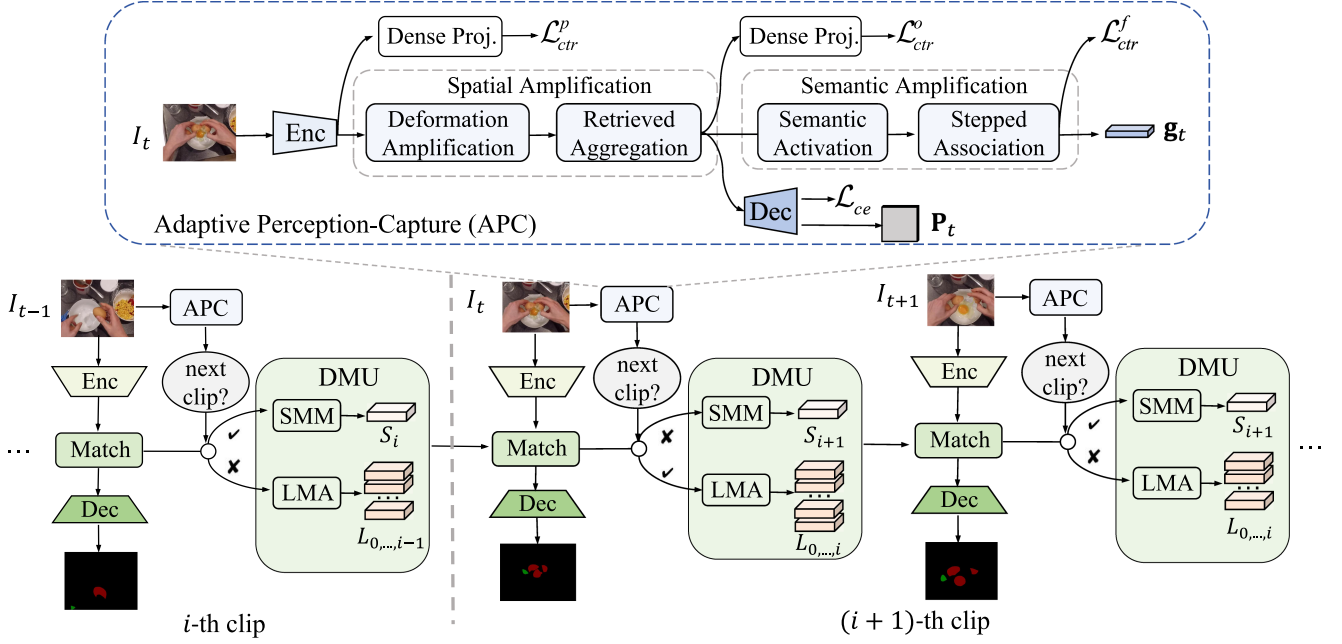


Fig. 2. Overview of our proposed Change-Sensitive Network (CSNet). S, L denotes short-term and long-term memory. Apart from matching with stored memory to generate the predicted mask, our proposed Adaptive Perception-Capture (APC) module generates the probability distribution P_t and global vector g_t for measuring the extent of spatial deformation and semantic transformation, which is used to determine whether to start the next clip or not, then APC activates different Dynamic Memory Update (DMU) modules including the Short-term Memory Modulation (SMM) module and the Long-term Memory Augmentation (LMA) module to dynamically update short-term and long-term memory respectively.

where $\langle \cdot, \cdot \rangle$ denotes the affinity calculation approaches [15], [16], [17], and the output V_t^q is decoded to produce the predicted mask $\hat{Y}_t$. Finally, the memory M_t is updated as

$$M_t = f(I_t, \hat{Y}_t, M_{t-\delta}), \quad (2)$$

where f denotes memory update mechanism [16], [17], [18], [19], [20], [23], [66], and δ denotes memory update interval.

Previous memory-based methods [16], [17], [18], [19], [20] usually adopt a fixed memory update interval δ . For example, DeAOT [18] update its short-term and long-term memory by setting δ to 1 and 5 respectively. However, when encountering complex scenarios, this fixed-interval memory update mechanism becomes fragile due to passively acquiring object-related features in two aspects: 1) When the target objects undergo significant semantic or spatial changes, the fixed-interval update of long-term memory may miss the object-changed features, leading to the matching process in (1) mainly based on the before-change features. 2) When there are insignificant changes of objects in an interval, the object-related features extracted from the multiple frames in this interval can be used to more adequately update short-term memory compared with only previous frame. To deal with these issues, we design an adaptive memory update mechanism in our proposed Change-Sensitive Network to dynamically update the short-term and long-term memory based on the extent of object semantic-spatial changes.

B. Network Overview

The overview of our proposed Change-Sensitive Network (CSNet) are shown in Fig. 2 which mainly consists of three components: 1) A basic matching process illustrated in (1) that enhances the object-specific features in the current frame via

the stored features in the memory. 2) An Adaptive Perception-Capture (APC) module to determine when to update memory by measuring the extent of object semantic transformations and spatial deformations, and the entire videos are divided into clips based on the memory update moments. 3) Dynamic Memory Update (DMU) modules to redefine how to update memory after dividing clips, including a Short-term Memory Modulation (SMM) module for aggregating expressive representations within clips, and a Long-term Memory Augmentation (LMA) module for highlighting object variations across clips.

Specifically, given the t -th video frame, denoted as I_t , the network sequentially encodes, matches and decodes it to generate the predicted mask. Meanwhile, the input frame I_t is fed into the APC module consisting of spatial and semantic amplifications to generate probability distribution P_t and global vector g_t , which are used to distinguish the extent of spatial deformation and semantic transformation respectively, thereby the APC module can determine whether to start the next clip or not. Then the current features enhanced by the matching module are updated into the memory by activating different dynamic memory update modules (i.e., SMM and LMA). When entering into a new clip, for example, a complete egg in the frame I_{t-1} is transformed to a broken egg in the frame I_t in Fig. 2, the Long-term Memory Augmentation (LMA) is activated to highlight object variations and append variation-enhanced prototypes L_t of the i -th clip in the long-term memory, and the short-term memory is reinitialized as S_{i+1} by the object-specific features of the current frame I_t . When the target objects undergo relatively smooth changes (from I_t to I_{t+1}), the Short-term Memory Modulation (SMM) module is activated to modulate object prototypes within the $(i+1)$ -th clip to update the prototypes S_{i+1} stored in the short

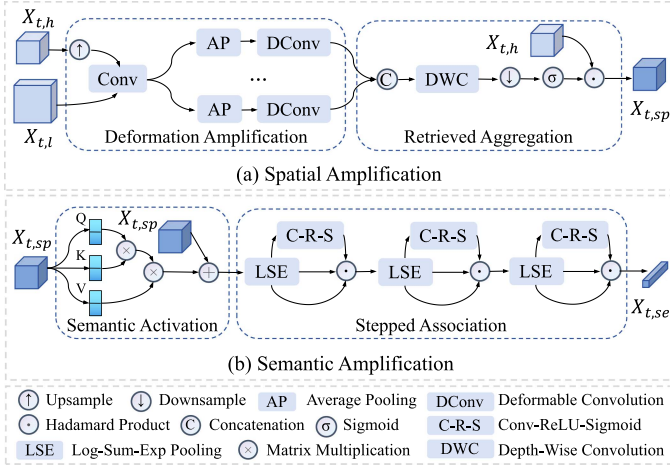


Fig. 3. Components of (a) Spatial Amplification and (b) Semantic Amplification in our proposed Adaptive Perception-Capture module.

term memory. Based on our proposed APC and DMU modules, our CSNet can adaptively update the memory according to the extent of object spatial-semantic changes, thus the object-related features can be effectively enhanced when matching with the change-aware and adaptively-updated memory.

C. Adaptive Perception-Capture

To proactively adjust memory according to object changes, we design an Adaptive Perception-Capture (APC) module to measure the extent of object semantic transformations and spatial deformations. This requires not only that the APC module can generate effective representations of video frames for space-time correspondence, but also the representations highlight as much semantic and spatial information of target objects as possible. Therefore, we delicately design Spatial and Semantic Amplifications to enhance the spatial and semantic informations respectively as shown in the upper part of Fig. 2. We combine our proposed Hierarchical Contrastive Learning at Frame, Object and Pixel levels with supervised learning in a one-stage deep-supervision network [73] during training. With the above design, our APC module can master the nuances of object semantic-spatial changes and discern temporal patterns of target objects.

For the t -th input frame I_t , the APC module outputs the probability distribution $\mathbf{P}_t$ and the global vector $\mathbf{g}_t$. These are used to measure the extent of object changes at spatial and semantic aspects respectively, by comparing them with the initial outputs $(\mathbf{P}_i, \mathbf{g}_i)$ of the current i -th clip. The measurements, including distribution consistency for spatial deformation ($\mathbf{P}_t$ vs. $\mathbf{P}_i$) and vector similarity for semantic transformation ($\mathbf{g}_t$ vs. $\mathbf{g}_i$), jointly determine whether to update memory and enter the $(i+1)$ -th clip.

Spatial Amplification: We perform spatial amplification on the encoded features $X_{t,h} \in \mathbb{R}^{C \times H \times W}$ and $X_{t,l} \in \mathbb{R}^{\frac{C}{2} \times 2H \times 2^W}$ of the fourth and third encoding stages. As shown in Fig. 3(a), we firstly upsample the feature $X_{t,h} \in \mathbb{R}^{C \times H \times W}$ to aggregate with the feature $X_{t,l} \in \mathbb{R}^{\frac{C}{2} \times 2H \times 2^W}$. As both average pooling (AP) layers and deformable convolution (DConv) layers are effective

to adapt to the geometric deformations of objects, we perform the deformation amplification on the aggregated feature with four parallel multi-scale branches with AP and DConv layers to produce $X_{t,s}^j \in \mathbb{R}^{\frac{C}{2} \times 2H \times 2^W}$, $j \in \{1, 2, 3, 4\}$. To retrieve spatial informations, the concatenated features are passed through depth-wise convolution layers and downsampled, activated, to generate the weight $W_t \in \mathbb{R}^{C \times H \times W}$. Finally, the weight is re-multiplied with the feature $X_{t,h}$ for producing the spatial amplification feature $X_{t,sp} \in \mathbb{R}^{C \times H \times W}$ as:

$$X_{t,sp} = W_t \odot X_{t,h}, \quad (3)$$

where $\odot$ denotes the hadamard product. The feature $X_{t,sp}$ is used for two purposes: (1) It is sent into decoder to generate the output probability distribution $\mathbf{P}_t$. (2) It is projected via 1×1 convolution layers to produce dense embeddings for object-level contrastive learning loss $\mathcal{L}_{ctr}^o$.

Semantic Amplification: Furthermore, we perform semantic amplification on the output feature $X_{t,sp}$ of spatial amplification. As shown in Fig. 3(b), we firstly perform preliminary semantic activation by embedding the feature $X_{t,sp}$ into query $X_{t,sp}^q \in \mathbb{R}^{C \times HW}$, key $X_{t,sp}^k \in \mathbb{R}^{HW \times C}$ and value $X_{t,sp}^v \in \mathbb{R}^{C \times HW}$ to obtain $X_{t,se} \in \mathbb{R}^{C \times H \times W}$ as:

$$X_{t,se} = \text{Softmax} \left(\frac{X_{t,sp}^q X_{t,sp}^k}{\sqrt{HW}} \right) X_{t,sp}^v + X_{t,sp}. \quad (4)$$

Then stepped association is performed by gradually down-sampling $X_{t,se}$ using log-sum-exp pooling (LSE) with scopes s^j , $j \in \{1, 2, 3\}$ set to $\frac{H}{8} \times \frac{W}{8}$, 4×4 , 2×2 . Then the features $X_{t,se}^j$ are sent into Convolution-ReLU-Sigmoid (CRS) layers to estimate association weights which are used to re-weight the feature $X_{t,se}^j$. The stepped association process can be formulated as:

$$X_{t,se}^j = \text{CRS} \left(\text{LSE}_{s^j} (X_{t,se}^{j-1}) \right) \odot \text{LSE}_{s^j} (X_{t,se}^{j-1}). \quad (5)$$

The final output $X_{t,se}^3 \in \mathbb{R}^{C \times 1 \times 1}$ is projected with a multi-layer perception (MLP) to produce the global vector $\mathbf{g}_t$ for the frame-level contrastive learning loss $\mathcal{L}_{ctr}^f$.

Frame-level Contrastive Learning: We firstly design frame-level contrastive learning loss to supervise the global vector and learn the effective representations for space-time correspondence. Formally, given a video with T frames $\mathcal{I} = \{I_t\}_{t=1}^T$, we evenly divide the entire video into K disjoint segments and randomly select one frame from each segment. For each sampled frame, we need to clarify its positive/negative (pos/neg) pairs for contrastive learning. Therefore, we firstly initialize the encoder in APC with the pretrained weights [74] and encode these K sampled frames to generate preliminary features $X \in \mathbb{R}^{C \times H \times W}$. Then we convert the features X to global vectors $\tilde{\mathbf{g}}$ after balancing the semantic contribution of objects and background with global weighted pooling as:

$$\tilde{\mathbf{g}}[c] = \frac{n_{\mathcal{B}}}{HW n_{\mathcal{O}}} \sum_{(h,w) \in \mathcal{O}} X[c, h, w] + \frac{n_{\mathcal{O}}}{HW n_{\mathcal{B}}} \sum_{(h,w) \in \mathcal{B}} X[c, h, w], \quad (6)$$

where $\mathcal{O}, \mathcal{B}$ denote the object and background region based on the ground-truth masks, and $n_{\mathcal{O}}, n_{\mathcal{B}}$ denote the number of object and background pixels in $\mathcal{O}, \mathcal{B}$ respectively. Finally, we treat each global vector as the anchor, denoted as $\tilde{\mathbf{g}}_{\alpha}$, and regard

the most/least similar vectors as pos/neg pairs by calculating the similarity score between each anchor and the remaining $K - 1$ vectors as:

$$\beta = \arg \max_{j \neq \alpha, j \in [1, K]} (\tilde{g}_\alpha \cdot \tilde{g}_j), \gamma = \arg \min_{j \neq \alpha, j \in [1, K]} (\tilde{g}_\alpha \cdot \tilde{g}_j), \quad (7)$$

where each anchor frame I_α forms pos/neg pairs with I_β, I_γ respectively. Besides, we extraly sample other K frames from other videos to form the negative bank $\mathcal{N}$ with the frame I_γ . After clarifying pos/neg pairs, these $2K$ frames are encoded, enhanced with spatial and semantic amplifications and projected to obtain the final global vector $g_j, j \in \{1, \dots, 2K\}$. We adopt InfoNCE [75] to calculate the frame-level contrastive learning loss as:

$$\mathcal{L}_{ctr}^f = -\log \frac{\exp(g_\alpha \cdot g_\beta / \tau)}{\exp(g_\alpha \cdot g_\beta / \tau) + \sum_{j \in \mathcal{N}} \exp(g_\alpha \cdot g_j / \tau)}, \quad (8)$$

where τ is a temperature hyper-parameter.

Object-level Contrastive Learning: To focus on the target objects in frames, we design object-level contrastive learning loss. Specifically, we firstly treat the objects in pos/neg frame pairs as pos/neg object pairs and regard the objects in other frames as negative object pairs to the object in the anchor frame. Then we perform the average pooling on the object area $\mathcal{O}$ in the anchor frame I_α , the positive sample I_β and the negative samples $\mathcal{N}$ to generate object-level feature vectors o, o^+ and $o^- \in \mathcal{N}_o$ respectively. Finally, we can calculate the object-level contrastive learning loss as:

$$\mathcal{L}_{ctr}^o = -\log \frac{\exp(o \cdot o^+ / \tau)}{\exp(o \cdot o^+ / \tau) + \sum_{o^- \in \mathcal{N}_o} \exp(o \cdot o^- / \tau)}. \quad (9)$$

Pixel-level Contrastive Learning: To understand the relationship between pixels in frames [76], we design pixel-level contrastive learning loss to enhance the pixel-level representations of target objects. Specifically, given a pixel-level feature vector p of the anchor frame I_α , we find the most similar pixel index s in the pos/neg frames I_β, I_γ as:

$$s = \arg \max_{j \in I_\beta \cup I_\gamma} (p \cdot p_j), \quad (10)$$

then we regard the pixel-level feature vector p_s as the positive sample p^+ and the remaining feature vectors in $I_\beta \cup I_\gamma$ constitute the negative samples p^- in $\mathcal{N}_p$. Finally, we can calculate the pixel-level contrastive learning loss as:

$$\mathcal{L}_{ctr}^p = \frac{1}{HW} \sum_{HW} -\log \frac{\exp(p \cdot p^+ / \tau)}{\exp(p \cdot p^+ / \tau) + \sum_{p^- \in \mathcal{N}_p} \exp(p \cdot p^- / \tau)}. \quad (11)$$

Therefore, our proposed frame-object-pixel **Hierarchical Contrastive Learning** loss can be formulated as:

$$\mathcal{L}_{ctr} = \mathcal{L}_{ctr}^f + \lambda \mathcal{L}_{ctr}^o + \mu \mathcal{L}_{ctr}^p, \quad (12)$$

where λ, μ are corresponding weighted coefficients.

Clip Division: During online inference, we firstly calculate the unique thresholds for each input video, then determine whether to enter into a new clip or not after comparing consistency and similarity scores with the thresholds.

Specifically, we first randomly sample K frames from the equally divided disjoint segments of the current input video.

Then we use the APC module to generate the probability distribution P_j and global vectors g_j of each sampled frames, and calculate the distribution consistency σ_{sp} and vector similarity σ_{se} of the two frames in sequence according to the order of the frames in the video as:

$$\sigma_{sp} = \text{JS}(P_j || P_{j+1}), \sigma_{sp} \in [0, 1], \quad (13)$$

$$\sigma_{se} = \frac{g_j \cdot g_{j+1}}{\|g_j\| \|g_{j+1}\|}, \sigma_{se} \in [0, 1], \quad (14)$$

where $j \in \{1, \dots, K - 1\}$, JS denotes Jensen-Shannon divergence function, and the closer output σ_{sp} is to 0, the more consistent the probability distribution is. Then we calculate the average of σ_{sp}^j and σ_{se}^j respectively to obtain average consistency $\bar{\sigma}_{sp}$ and average similarity $\bar{\sigma}_{se}$ as the thresholds of the current input video.

After clarifying the unique thresholds of each video, we start to segment the target objects in the current input video. Specifically, we set the first frame as the anchor frame of the first clip, and extract the anchor probability distribution P_i and global vector g_i of the first clip where $i = 1$. Then for the following input frame I_t , we calculate the distribution consistency σ_{sp} between P_t and P_i , and the vector similarity σ_{se} between g_t and g_i . We compare them with the thresholds respectively and enter the next $(i + 1)$ -th clip when σ_{sp} is higher than the threshold $\bar{\sigma}_{sp}$ or σ_{se} is lower than the threshold $\bar{\sigma}_{se}$. Then we set the current input frame as the anchor frame of the new clip, update the anchor probability distribution P_{i+1} and global vector g_{i+1} and continue the comparison until the target objects in the entire video are segmented.

D. Dynamic Memory Update

Once separating the entire video into clips based on the extent of object semantic-spatial changes, it is necessary to dynamically manage the memory within and across clips to adapt memory to object changes. To this end, we design 1) a Short-term Memory Modulation module to modulate object similarities and extract pivotal informations within clips, 2) a Long-term Memory Augmentation module to perceive object variations and enhance crucial differences across clips. Through considering both intra-clip and inter-clip object changes, we can enable smooth prototype updates in short-term memory and adaptive retention of variation-enhanced prototypes in long-term memory.

Short-term Memory Modulation (SMM): Compared with taking object features of the previous frame as short-term memory [18], [20], or using GRU [72] to continuously update short-term memory throughout the entire video [17], [34], SMM extracts the prototypes of target objects within each clip where changes are insignificant. SMM consists of two components: similarity-weighted modulation and short-term memory propagation as shown in Fig. 4(a).

Similarity-Weighted Modulation: (Fig. 5(a)) extracts object prototypes within the current i -th clip. Formally, given the stored probability distribution P_i^j and global vector g_i^j of the j -th frame in the i -th clip, where $j \in \{1, \dots, C_i\}$ and C_i denotes the current number of frames in the i -th clip, we compute the distribution consistency (σ_{sp}^j between P_i^j and $P_i^{C_i}$) and vector similarity (σ_{se}^j

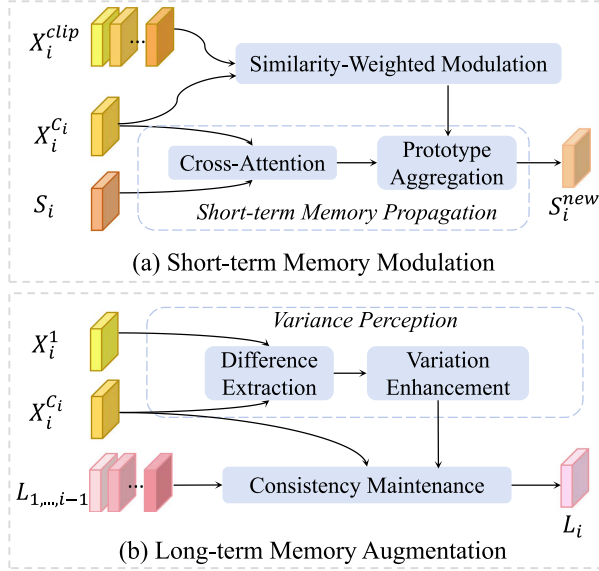


Fig. 4. Composition of our proposed dynamic memory update modules including (a) Short-term Memory Modulation module and (b) Long-term Memory Augmentation module. X_i^{clip} denotes all object-related feature $X_i^j, j \in \{1, \dots, C_i - 1\}$ of all frames in the current i -th clip where C_i denotes the number of frames in the i -th clip, and $X_i^{C_i}$ denotes the current object-related feature of the i -th clip which is generated following [17], [19]. $S_i, L_{1, \dots, i-1}$ denote the features stored in short-term and long-term memory. S_i^{new}, L_i denote the updated short-term memory and the generated long-term memory of the i -th clip.

between g_i^j and $g_i^{C_i}$), averaging them to get the similarity weight σ_i^j . These weights for $j \in \{1, \dots, C_i - 1\}$ are normalized via Softmax to obtain the final weight $\tilde{\sigma}_i^j$. Then we perform cross-attention between each stored feature $X_i^j \in \mathbb{R}^{HW \times C}$ and the current feature $X_i^{C_i} \in \mathbb{R}^{C \times HW}$:

$$\tilde{X}_i^j = \text{Softmax} \left(\frac{X_i^j X_i^{C_i}}{\sqrt{C}} \right) X_i^j. \quad (15)$$

Finally, we modulate these outputs $\tilde{X}_i^j$ based on the normalized similarity weights $\tilde{\sigma}_i^j$ to generate the object prototypes $\tilde{X}_i^{C_i}$ of the current feature $X_i^{C_i}$ as:

$$\tilde{X}_i^{C_i} = \sum_j^{C_i-1} \tilde{\sigma}_i^j \tilde{X}_i^j. \quad (16)$$

Short-term Memory Propagation generates the final updated short-term memory S_i^{new} . We first update the previous short-term memory $S_i \in \mathbb{R}^{HW \times C}$ using cross-attention with the current feature $X_i^{C_i} \in \mathbb{R}^{C \times HW}$:

$$\tilde{S}_i = \text{Softmax} \left(\frac{X_i^{C_i} S_i^T}{\sqrt{C}} \right) X_i^{C_i} + S_i, \quad (17)$$

where S_i^T denotes the transpose of S_i . Finally, the result $\tilde{S}_i$ is concatenated with the modulated prototypes $\tilde{X}_i^{C_i}$ and passed through a depth-wise convolution to produce $S_i^{new} \in \mathbb{R}^{H \times W \times C}$, which serves as the new short-term memory for the next frame.

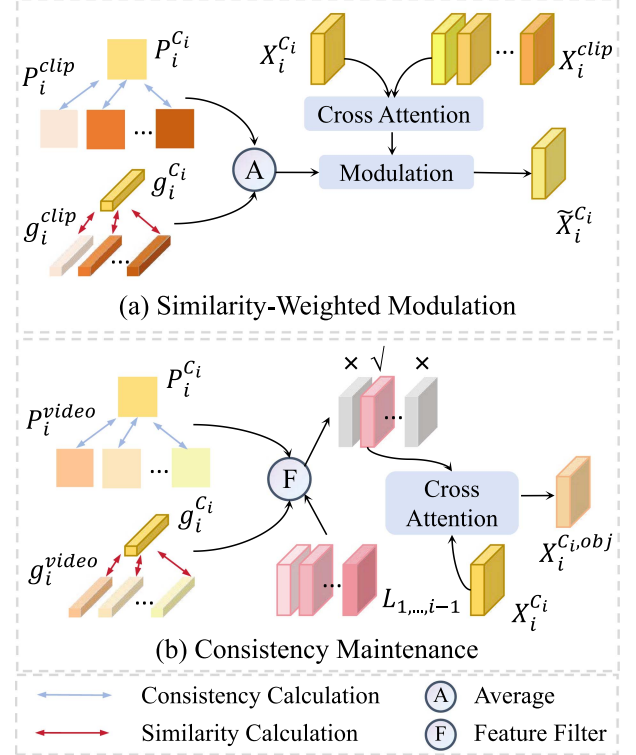


Fig. 5. Structure of (a) Similarity-Weighted Modulation and (b) Consistency Maintenance in our proposed Short-term Memory Modulation module and Long-term Memory Augmentation module respectively. P_i^{clip} and g_i^{clip} denote the distributions and vectors within the i -th clip. P_i^{video} and g_i^{video} denote the distributions and vectors of key frames in previous clips across the entire video.

The proposed Short-term Memory Modulation module takes into account the pivotal features of target objects within clips, while avoiding aggregating redundant features. Compared to previous-frame update [18], [20], our design updates short-term memory with clip-wise informations about target objects. Meanwhile, compared to entire-video update [17], [19], switching to the next clip based on the object changes can avoid incorporating noises into short-term memory that mismatch the current target object.

Long-term Memory Augmentation (LMA): As the APC module divides the video into clips, LMA is designed to explicitly perceive and store target object variations across clips when entering a new clip. As shown in Fig. 4(b), LMA consists of two components: variance perception and consistency maintenance.

Variance Perception extracts object variations within the current i -th clip. The object variation X_i^{var} is extracted by comparing the current feature $X_i^{C_i} \in \mathbb{R}^{HW \times C}$ with the first frame's feature $X_i^1 \in \mathbb{R}^{C \times HW}$ of the clip:

$$X_i^{var} = X_i^{C_i} - \text{Softmax} \left(\frac{X_i^{C_i} X_i^1}{\sqrt{C}} \right) X_i^{C_i}. \quad (18)$$

Then we enhance the variations of $X_i^{C_i}$ after aggregating (Agg) the output feature X_i^{var} and the current feature $X_i^{C_i}$ with depth-wise convolution layers as:

$$X_i^{C_i, var} = X_i^{C_i} \odot \text{Softmax} \left(\text{Agg} \left(X_i^{var}, X_i^{C_i} \right) \right). \quad (19)$$

Consistency Maintenance Fig. 5(b) ensures that only relevant object prototypes are stored. Given stored outputs (P_k, g_k, X_k) of key frames (dividing frames of different clips) from the past L clips, we calculate σ_{sp}^k and σ_{se}^k between these stored values and the current frame's outputs ($P_i^{C_i}, g_i^{C_i}$). We filter for key frames where both σ_{sp}^k and σ_{se}^k are greater than the current clip's final consistency values ($\sigma_{sp}^{C_i}, \sigma_{se}^{C_i}$) from the APC module. The features ($X_k, k \in \{1, \dots, L'\}$) of these retained key frames are concatenated into $M_i \in \mathbb{R}^{L'HW \times C}$. We perform cross-attention between the current feature $X_i^{C_i} \in \mathbb{R}^{HW \times C}$ and the aggregated memory M_i to retrieve object consistency information:

$$X_i^{C_i, obj} = \text{Softmax} \left(\frac{X_i^{C_i} M_i^T}{\sqrt{C}} \right) M_i + X_i^{C_i}, \quad (20)$$

where M_i^T denotes the transpose of M_i . Finally, we generate the variation-enhanced features of target objects by aggregating features $X_i^{C_i, var}$ and $X_i^{C_i, obj}$, and reweighting the current feature $X_i^{C_i}$ as:

$$X_i^{C_i, long} = X_i^{C_i} \odot \text{Softmax} \left(\text{Agg} \left(X_i^{C_i, var}, X_i^{C_i, obj} \right) \right), \quad (21)$$

then we append the output $X_i^{C_i, long}$ in the long-term memory for the crucial object variations of the current clip. Compared with previous methods [17], [19] of compressing long-term memory, our proposed Long-term Memory Augmentation module retains the variant informations of object targets across clips, which can act as the object prototypes without compression. When the object changes are similar to the previous ones, we can effectively match and enhance the object-related features based on the stored variations in the long-term memory.

IV. EXPERIMENTS

In this section, we introduce the experimental settings, the quantitative comparison results on common, complex and long-term datasets between our proposed Change-Sensitive Network (CSNet) and existing state-of-the-art Semi-VOS methods, and the performance analysis to the current evaluation results through sufficient ablation studies.

A. Experimental Settings

We introduce the experiment settings including benchmark datasets, evaluation metrics, implementation and training details as follows.

Datasets: To comprehensively verify the effectiveness of our proposed CSNet, we evaluate our CSNet on common datasets including DAVIS 2016 [33], DAVIS 2017 [26], YouTube-VOS 2018 [27], YouTube-VOS 2019 [27], emerging complex datasets including VOST [29] and MOSE [30], and long-video datasets LVOS [34], BURST [35]. As one of emerging complex datasets, VOST mainly focuses on object transformations where the color, texture and shape of objects can be dramatically changed. Meanwhile, MOSE pays more attention to complex scenarios including the long-term disappearance-reappearance of objects, heavy occlusions, etc. In comparison, the complexity in common datasets including DAVIS and YouTube-VOS is far less than that in complex datasets. The validation set of

DAVIS 2016 only has 20 video sequence with the shortest video length and the validation set of DAVIS 2017 contains 30 videos with multi-object annotations. YouTube-VOS 2018 contains 3471 videos for training (65 categories) and 507 videos for validation, and YouTube-VOS 2019 contains additional 26 unseen categories to evaluate the generalization of the VOS methods. The target objects in these common datasets are usually salient and dominant, and the changes of scenes and objects in these videos are not significant. The long-term videos in LVOS dataset have an average duration of 1.59 minutes which are 20 times longer than short-term videos in these common and complex datasets. Moreover, the long-term dataset BURST consists of 2914 long-term videos with average duration of 36.8 seconds, covering multiple types of scenes and 482 object classes.

Evaluation Metrics: For DAVIS [26], [33], MOSE [30] and LVOS [34] datasets, we follow previous methods [17], [30], [34] to adopt region similarity $\mathcal{J}$, contour accuracy $\mathcal{F}$ and their average $\mathcal{J} \& \mathcal{F}$ as the evaluation metrics. The metric $\mathcal{J}$ measures region similarity by calculating intersection-over-union (IoU) between the predicted mask and ground-truth mask which captures the fraction of pixels that are correctly labeled. The metric $\mathcal{F}$ measures contour accuracy which is how accurate the boundaries of predicted masks are. For YouTube-VOS [27] datasets, we report $\mathcal{J}$ and $\mathcal{F}$ for both seen and unseen categories, and the averaged overall score $\mathcal{G}$ following [16], [17]. For VOST [29] dataset, we report $\mathcal{J}$ and $\mathcal{J}_{tr}$ which represents the region similarity of the last 25% of the frames following [29]. For BURST [35] dataset, we report Higher Order Tracking Accuracy (HOTA) [81] on common and uncommon class subsets.

Implementation Details: In our proposed CSNet, we adopt ResNet-50 as the encoder and the same decoder following [17], [19]. In our proposed APC module, we adopt the lightweight backbone ResNet-18 [82] as the encoder, and the decoder only consists of upsampling blocks with merging features through skip-connections. Each frame are resized to 256×256 before entering our APC module for higher efficiency during training and inference. The spatial amplification consists of four parallel branches, and each branch contains an average pooling layer and a deformable convolution layer. The filter size of average pooling layers in four branches are $1 \times 1, 3 \times 3, 5 \times 5, 7 \times 7$ respectively, and we set the stride to 1 and the paddings to 0, 1, 2, 3 to keep the spatial size the same in different branches. After increasing the receptive field through average pooling layers, we reduce the filter size of the deformable convolution to ensure the effectiveness of the parallel branch.

Training Details: We implement our CSNet with PyTorch [83] and adopt four 48 GB A6000 GPUs to train it. Following previous methods [16], [17], [18], we firstly pretrain our CSNet on static image datasets [84], [85], [86], [87], [88]. Then we fine-tune the pretrained CSNet on video datasets DAVIS 2017 [26] and Youtube-VOS 2018 [27] to validate the performance on these common datasets. Meanwhile, we fine-tune the pretrained CSNet on VOST and MOSE datasets respectively following the same training setting in [29], [30] to verify the performance on complex datasets. Finally, we fine-tune the pretrained CSNet on LVOS [34] dataset following the same training setting [34] to

TABLE I

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART SEMI-VOS METHODS ON COMMON DATASETS INCLUDING DAVIS 2017 TEST, DAVIS 2017 VAL AND YOUTUBE-VOS 2019 VAL. WE RECORD THE INFERENCE SPEED (FPS) ON THE SAME A6000 GPU DEVICE AND * DENOTES THAT WE RETEST THE SPEED.

Method	Pub.	Backbone	DAVIS 2017 test			DAVIS 2017 val			YouTube-VOS 2019 val					FPS
			$\mathcal{J}\&\mathcal{F}\uparrow$	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	$\mathcal{J}\&\mathcal{F}\uparrow$	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	$\mathcal{G}\uparrow$	$\mathcal{J}_s\uparrow$	$\mathcal{F}_s\uparrow$	$\mathcal{J}_u\uparrow$	$\mathcal{F}_u\uparrow$	
CFBI [12]	ECCV ₂₀	ResNet-50	75.0	71.4	78.7	81.9	79.3	84.5	81.6	80.2	84.6	77.2	84.5	5.9
RMNet [77]	CVPR ₂₁	ResNet-50	75.0	71.9	78.1	83.5	81.0	86.0	-	-	-	-	-	12
CFBI+ [49]	TPAMI ₂₁	ResNet-50	75.6	71.6	79.6	82.9	90.1	85.7	82.9	80.6	85.2	78.9	86.8	5.6
STCN [16]	NIPS ₂₁	ResNet-50	76.1	72.7	79.6	85.4	82.2	88.6	82.7	81.1	85.4	78.2	85.9	20.2
AOT [20]	NIPS ₂₁	ResNet-50	79.6	75.9	83.3	84.9	82.3	87.5	84.1	83.5	88.1	78.4	86.3	18
BATMAN [78]	ECCV ₂₂	ResNet-50	82.2	78.4	86.1	86.2	83.2	89.3	85.0	84.5	89.3	79.0	87.2	-
RDE [23]	CVPR ₂₂	ResNet-50	77.4	73.6	81.2	84.2	80.8	87.5	81.9	81.1	85.5	76.2	84.8	27
XMem [17]	ECCV ₂₂	ResNet-50	81.0	77.4	84.5	86.2	82.9	89.5	85.5	84.3	88.6	80.3	88.6	26*
DeAOT [18]	NIPS ₂₂	ResNet-50	80.7	76.9	84.5	85.2	82.2	88.2	85.9	84.6	89.4	80.8	88.9	27*
KMN [53]	TPAMI ₂₂	ResNet-50	78.1	75.1	81.0	83.5	80.9	86.1	-	-	-	-	-	7.7
ISVOS [67]	CVPR ₂₃	ResNet-50	82.8	79.3	86.2	87.1	83.7	90.5	86.1	85.2	89.7	80.7	88.9	-
LLB [79]	AAAI ₂₃	ResNet-50	-	-	-	84.6	81.5	87.7	83.6	81.7	86.5	79.2	87.0	8.3
SimVOS [25]	ICCV ₂₃	ViT-L	82.3	78.7	85.8	88.5	85.4	91.5	84.2	83.1	-	79.1	-	2.0
RAVOS [22]	TIP ₂₄	ResNet-50	80.8	77.1	84.5	86.1	82.9	89.3	82.8	81.9	86.6	77.5	85.4	42
Cutie [19]	CVPR ₂₄	ResNet-50	84.2	80.6	87.7	88.8	85.4	92.3	86.1	85.5	90.0	80.6	88.3	36*
AOST [66]	TPAMI ₂₄	Swin-B	82.7	78.8	86.6	86.3	83.1	89.4	86.0	85.0	89.8	80.3	88.8	11*
STMA [80]	ECCV ₂₄	ConvViT-B	87.1	83.7	90.5	90.4	87.3	93.5	86.8	85.6	90.0	81.8	89.5	14*
CSNet (Ours)	-	ResNet-50	87.4	84.1	90.7	90.6	87.6	93.7	87.2	86.2	90.3	82.4	89.9	31

TABLE II

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART SEMI-VOS METHODS ON COMMON DATASETS INCLUDING THE VALIDATION SETS OF DAVIS 2016 AND YOUTUBE-VOS 2018

Method	DAVIS 2016 val			YouTube-VOS 2018 val				
	$\mathcal{J}\&\mathcal{F}\uparrow$	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	$\mathcal{G}\uparrow$	$\mathcal{J}_s\uparrow$	$\mathcal{F}_s\uparrow$	$\mathcal{J}_u\uparrow$	$\mathcal{F}_u\uparrow$
CFBI [12]	89.4	88.3	90.5	81.8	81.9	86.3	75.6	83.4
RMNet [77]	88.8	88.9	88.7	81.5	82.1	85.7	75.7	82.4
CFBI+ [49]	89.9	88.7	91.1	82.8	81.8	86.6	77.1	85.6
STCN [16]	91.6	90.8	92.5	83.0	81.9	86.5	77.9	85.7
AOT [20]	91.1	90.1	92.1	84.1	83.7	88.5	78.1	86.1
BATMAN [78]	92.5	90.7	94.2	85.3	84.7	89.8	79.2	87.4
RDE [23]	91.1	89.7	92.5	-	-	-	-	-
XMem [17]	91.5	90.4	92.7	85.7	84.6	89.3	80.2	88.7
DeAOT [18]	92.3	90.5	94.0	86.0	84.9	89.9	80.4	88.7
KMN [53]	91.2	90.2	92.1	82.7	81.9	77.5	86.0	85.1
ISVOS [67]	92.6	91.5	93.7	86.3	85.5	90.2	80.5	88.8
LLB [79]	-	-	-	83.8	82.1	87.0	79.1	87.0
SimVOS [25]	93.6	92.0	95.3	-	-	-	-	-
RAVOS [22]	91.7	90.8	92.6	83.2	82.2	86.9	77.9	85.9
Cutie [19]	-	-	-	86.1	85.8	90.5	80.0	88.0
AOST [66]	92.4	90.6	94.1	-	-	-	-	-
STMA [80]	93.4	91.9	94.8	86.4	85.0	89.6	81.5	89.3
CSNet (Ours)	93.5	92.1	94.9	87.0	85.6	89.9	82.1	90.4

verify the performance when dealing with long-term videos. When training our proposed CSNet, we use AdamW as the optimizer, combine the BCE loss [13], soft Jaccard loss [89] and our proposed hierarchical contrastive learning loss with equal weight. The weighted coefficients λ and μ in (12) are fixed as 2.0 and 1.0 for optimal performance. We use AdamW [90] as the optimizer with a learning rate of 1e-4, batch size of 32, a weight decay of 0.001. We set 100 K iterations with no learning rate decay during pre-training. During training on common datasets in main-training stage, we set 150 K iterations with learning rate reduced by 10 times after 120 K iterations. During fine-tuning on complex or long-term datasets in main-training stage, we set 100 K iterations with learning rate reduced by 10 times after 80 K iterations.

B. Comparison With State-of-The-Art

We conduct extensive and comprehensive experiments to evaluate our proposed CSNet on common, complex and long-term datasets, and compare the evaluation results with the state-of-the-art Semi-VOS methods including CFBI [12], RMNet [77], CFBI+ [49], STCN [16], AOT [20], BATMAN [78], RDE [23], XMem [17], DeAOT [18], KMN [53], ISVOS [67], LLB [79], RAVOS [22], Cutie [19], AOST [66] and STMA [80]. As the Semi-VOS methods evaluated in the benchmark of the emerging complex and long-term datasets are relatively few, to more sufficiently validate the effectiveness of our proposed CSNet when evaluated on the complex and long-term datasets, we retrain more latest SOTA Semi-VOS methods on these datasets under the same setting [29], [30], [34].

Comparison on common datasets: We conduct evaluation experiments on common datasets including the validation set of DAVIS 2016 [33], the validation and test set of DAVIS 2017 [26], the validation sets of YouTube-VOS 2018 [27] and YouTube-VOS 2019 [27]. The experimental results are shown in

Tables I and II. Compared to existing SOTA Semi-VOS methods, our proposed CSNet can achieve the highest accuracy and the real-time efficiency when evaluated on common datasets. Compared with the latest SOTA method Cutie [19] with the same backbone ResNet-50 [82], our proposed CSNet can achieve 3.2%, 1.8% improvements in terms of $\mathcal{J}\&\mathcal{F}$ on DAVIS 2017 test and validation sets, and 1.1%, 0.9% improvements in terms of $\mathcal{G}$ on YouTubeVOS 2019 and YouTubeVOS 2018 validation sets. Although the STMA method [80] adopts a stronger backbone ConvViT-B [91], our proposed CSNet can still lead in accuracy and be twice as fast in segmentation efficiency. Therefore, our proposed CSNet can effectively and efficiently segment target objects in these common datasets after learning when and how to update memory.

TABLE III

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART SEMI-VOS METHODS ON COMPLEX DATASETS INCLUDING VOST VAL, VOST TEST AND MOSE VAL

Method	VOST val		VOST test		MOSE val		
	$\mathcal{J} \uparrow$	$\mathcal{J}_{tr} \uparrow$	$\mathcal{J} \uparrow$	$\mathcal{J}_{tr} \uparrow$	$\mathcal{J} \uparrow$	$\mathcal{F} \uparrow$	$\mathcal{F} \uparrow$
CFBI [12]	45.0	32.0	43.9	32.1	45.7	41.2	50.1
CFBI+ [49]	46.0	32.6	46.7	31.6	47.8	42.9	52.7
AOT-R50 [20]	48.7	36.4	49.9	37.1	57.2	53.1	61.3
RDE [23]	41.3	30.6	40.7	30.2	48.8	44.6	52.9
XMem [17]	44.1	33.8	44.0	32.0	59.6	55.4	63.7
DeAOT-R50 [18]	50.4	36.1	50.3	36.2	64.1	59.5	68.7
LLB [79]	42.4	31.9	41.9	30.2	52.0	48.6	55.5
AOT+ [29]	49.4	38.2	49.1	38.0	57.8	54.2	61.4
Cutie [19]	47.5	32.9	46.9	32.6	68.3	64.2	72.3
RMem [21]	51.8	40.4	-	-	-	-	-
CSNet (Ours)	54.3	42.2	53.6	41.4	73.2	69.5	76.9

TABLE IV

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART SEMI-VOS METHODS ON LONG-VIDEO DATASET LVOS. MB DENOTES THE KIND OF MEMORY BANK. F+P, A, C, LT AND FIFO DENOTE FIRST AND PREVIOUS FRAME, ALL FRAMES, COMPRESSED MEMORY, LONG-TERM MEMORY AND FIRST-IN-FIRST-OUT MEMORY RESPECTIVELY. MEM(G): MAXIMUM GPU MEMORY USAGE.

Method	MB	LVOS val			LVOS test			Mem(G)
		$\mathcal{J} \uparrow$	$\mathcal{F} \uparrow$	$\mathcal{F} \uparrow$	$\mathcal{J} \uparrow$	$\mathcal{F} \uparrow$	$\mathcal{F} \uparrow$	
CFBI [12]	F+P	51.5	46.2	56.7	44.8	40.2	49.4	3.82
STCN [16]	A	48.9	43.9	54.0	48.3	44.0	52.5	0.92
RDE [23]	C	53.7	48.3	59.2	50.2	45.7	54.6	1.0
XMem [17]	LT	52.9	48.1	57.7	50.9	46.5	55.3	3.34
DDMemory [34]	C	61.9	56.3	67.4	55.7	50.3	61.2	0.88
Cutie [19]	LT	60.1	55.9	64.2	56.2	51.8	60.5	2.36
STMA [80]	-	-	-	-	57.4	53.3	61.5	-
CSNet (Ours)	FIFO	65.5	61.3	69.7	58.9	54.7	63.1	3.05

Comparison on complex datasets: We conduct evaluation experiments on complex datasets including the validation and test sets of VOST [29], the validation set of MOSE [30]. As shown in Table III, our proposed CSNet can achieve significant improvements on both VOST and MOSE datasets. The SOTA Semi-VOS methods such as RDE [23], XMem [17], Cutie [19] with the fixed-interval memory update mechanism suffer from significant accuracy declines when encountering complex object transformations and deformations. In comparison, our proposed CSNet demonstrates superior robustness to deal with complex scenarios. Through dynamically updating memory in response to object semantic-spatial changes, our proposed CSNet can achieve 2.5% in terms of $\mathcal{J}$ on the validation set of VOST and 4.9% in terms of $\mathcal{J} \& \mathcal{F}$ on the validation set of MOSE compared to RMem [21] and Cutie [19] respectively. The significant improvements of accuracy when evaluated on these complex datasets sufficiently illustrate the effectiveness of our proposed CSNet when addressing the intricate challenges in the emerging complex scenarios.

Comparison on long-term datasets: We conduct evaluation experiments on long-term datasets including the validation and test sets of LVOS [34] and BURST [35] datasets, and the results are shown in Table IV. We adopt the First-In-First-Out (FIFO) strategy to keep the memory constant, which can avoid excessive memory growth during inference on long-term videos with thousands of frames. Our proposed CSNet can outperform these existing SOTA Semi-VOS methods with carefully-designed

TABLE V

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART SEMI-VOS METHODS ON LONG-VIDEO DATASET BURST. MB DENOTES THE KIND OF MEMORY BANK.

Method	MB	Mem (G)	BURST val			BURST test		
			All	Com.	Unc.	All	Com.	Unc.
DeAOT [18]	FIFO	10.8	51.3	56.3	50.0	53.2	53.5	53.2
XMem [17]	FIFO	3.03	52.9	56.0	52.1	55.9	57.6	55.6
XMem [17]	LT	3.34	55.1	57.9	54.4	58.2	59.5	58.0
Cutie [19]	LT	2.36	58.4	61.8	57.5	62.6	63.8	62.3
CSNet (Ours)	FIFO	3.05	60.3	64.1	59.4	65.4	66.4	65.1

TABLE VI

ABLATION STUDIES ON THE IMPACT OF OUR PROPOSED KEY COMPONENTS TO THE EFFICIENCY, MEMORY USAGE AND ACCURACY. KEY COMPONENTS IN OUR METHOD INCLUDE ADAPTIVE PERCEPTION-CAPTURE (APC), SHORT-TERM MEMORY MODULATION (SMM), LONG-TERM MEMORY AUGMENTATION (LMA) AND HIERARCHICAL CONTRASTIVE LEARNING (HCL).

ID	Setting	FPS	Mem (G)	VOST val		MOSE val		
				$\mathcal{J} \uparrow$	$\mathcal{F} \uparrow$	$\mathcal{J} \uparrow$	$\mathcal{F} \uparrow$	$\mathcal{F} \uparrow$
1	w/o APC	28	2.86	47.9	34.7	65.4	61.7	69.1
2	w/o SMM	32	2.57	50.1	37.9	68.9	65.2	72.6
3	w/o LMA	34	2.49	51.3	38.8	70.1	67.0	73.2
4	w/o HCL	25	3.15	42.1	28.6	59.6	56.3	62.9
5	Ours	31	2.62	54.3	42.2	73.2	69.5	76.9

memory control strategies, for example, our CSNet achieve 1.5% accuracy improvements compared to STMA method in terms of $\mathcal{J} \& \mathcal{F}$ on LVOS test set, and achieves 2.8% improvements compared to Cutie method in terms of HOTA on BURST test set. These results demonstrate significant advantages of our proposed adaptive memory update mechanism which can effectively perceive and update the object prototypes of the key moments in the long-term videos.

Qualitative comparisons: We further show the qualitative comparisons between our proposed CSNet and SOTA method Cutie [19]. As shown in Fig. 6, compared to existing SOTA method Cutie, our method can achieve significant improvements when facing (a) dramatic semantic transformations, (b) severe spatial deformations, (c) object disappearance-reappearance, (d) heavy occlusion, (e) rapid motion, and (f) similar distractors. In Fig. 7, we show the segmentation results of the entire video. Our proposed CSNet can effectively address emerging challenges such as (b) similar distractors, (c-d) long-term object disappearance-reappearance and (e) dramatical deformation.

C. Performance Analysis

We conduct extensive ablation studies to validate the effect of each module in our proposed Change-Sensitive Network.

Effect of key components: We quantify the effect of each proposed component on efficiency, memory usage and performance. As shown in Table VI, we respectively remove components including Adaptive Perception-Capture (APC), Short-term Memory Modulation (SMM), Long-term Memory Augmentation (LMA) and Hierarchical Contrastive Learning (HCL), then record segmentation speed (FPS), maximum memory usage (Mem) and accuracy on validation sets of complex datasets [29], [30]. Without APC, our method can only update memory at

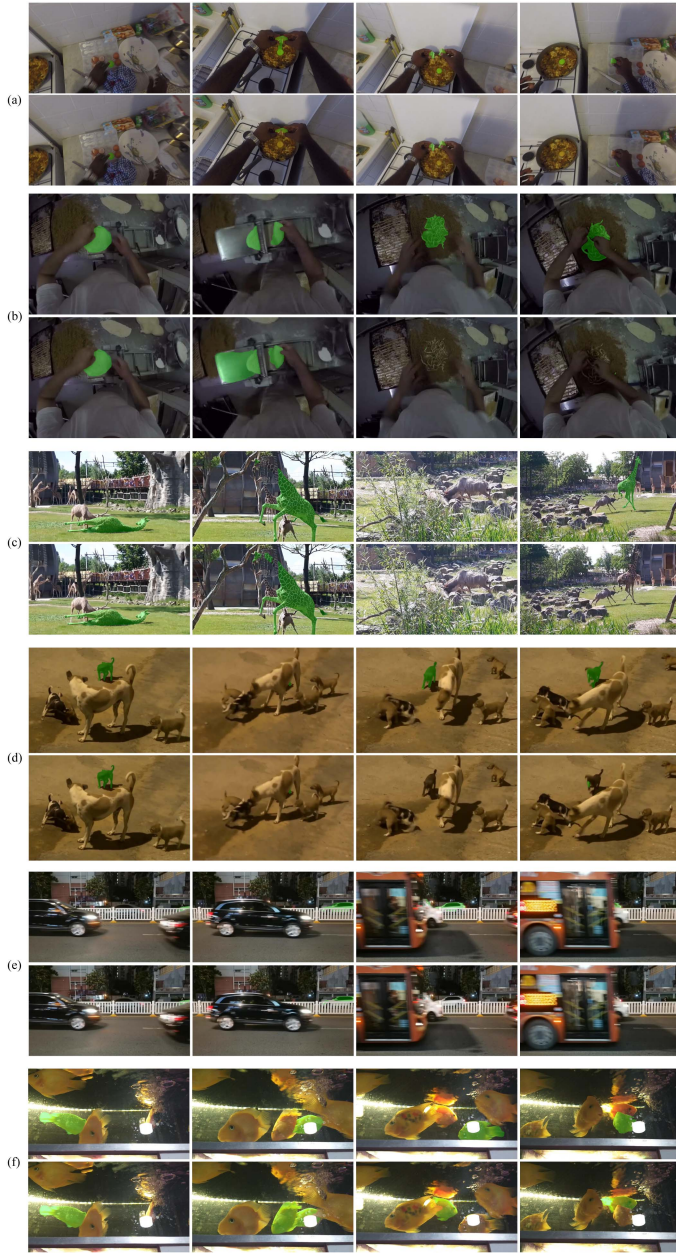


Fig. 6. Qualitative comparisons between our proposed CSNet (top) and SOTA method Cutie (bottom) when facing different scenarios including (a) dramatic semantic transformations, (b) severe spatial deformations, (c) object disappearance-reappearance, (d) heavy occlusion, (e) rapid motion, and (f) similar distractors.

fixed-interval, leading to redundant memory, efficiency and accuracy decrease in setting 1. Dynamic memory update modules including SMM and LMA can both effectively improve the accuracy (setting 2, 3). Moreover, without HCL, our method is sensitive to background changes which introduces noisy features in memory, significantly reducing efficiency and accuracy as shown in setting 4. In the following, we conduct more detailed ablation studies inside each component.

Effect of adaptive perception-capture module: To validate the effectiveness of our proposed Adaptive Perception-Capture

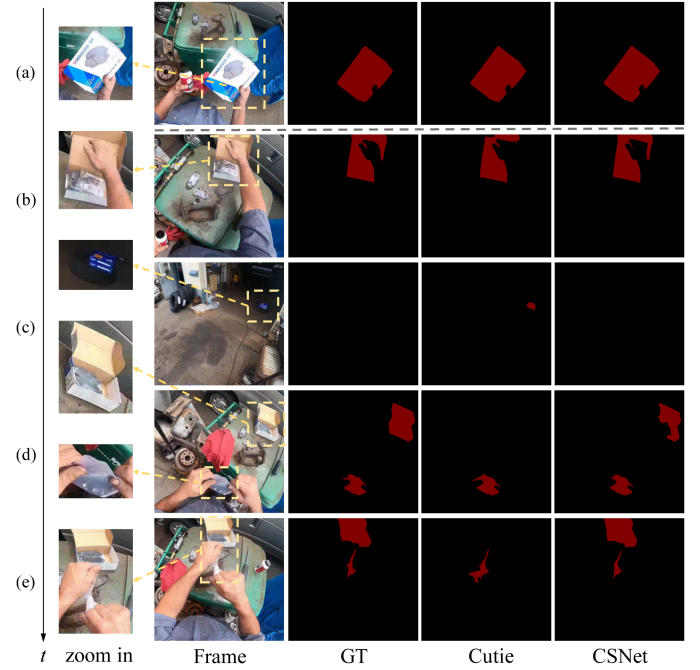


Fig. 7. Qualitative comparisons between existing SOTA method Cutie and our proposed CSNet. Given the first frame and groundtruth (gt) mask as (a), the segmentation results are shown when faced with (b) similar distractors, (c-d) long-term object disappearance-reappearance and (e) dramatical deformation.

TABLE VII
ABLATION STUDIES ON COMPLEX DATASETS VOST AND MOSE FOR OUR PROPOSED ADAPTIVE PERCEPTION-CAPTURE (APC) MODULE WHERE SpM AND SeM DENOTE THE MEASUREMENTS OF SPATIAL AND SEMANTIC CHANGES, SpA AND SeA DENOTE SPATIAL AND SEMANTIC AMPLIFICATION COMPONENTS RESPECTIVELY IN OUR PROPOSED APC MODULE

ID	Setting	VOST val		MOSE val		
		$\mathcal{J}$	$\mathcal{J}_{tr}$	$\mathcal{J} \& \mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$
1	w/o APC	47.9	34.7	65.4	61.7	69.1
2	w/o SpM	49.6	36.3	68.9	64.8	73.0
3	w/o SeM	50.7	39.2	71.2	66.8	75.6
4	w/o SpA	50.1	38.0	70.8	66.5	75.1
5	w/o SeA	52.2	39.8	71.5	67.2	75.8
6	Ours	54.3	42.2	73.2	69.5	76.9

(APC) module, we conduct corresponding ablation studies shown in Table VII. In setting 1, we directly adopt the fixed-interval memory update mechanism following existing representative methods [16], [17], [19], and the accuracy immediately drops to the level of these methods which sufficiently demonstrates the fixed-interval memory update mechanism cannot align with object changes and miss the crucial informations during object changes. Then we evaluate different measurements of spatial and semantic changes (SpM and SeM) in setting 2,3, and the results show that each measurement at different aspects is helpful to acquire the informations of object changes. Finally, we evaluate the impact of different components in our proposed APC module including Spatial Amplification (SpA) and Semantic Amplification (SeA) in setting 4,5. These components can improve 3.2% and 2.1% in terms of $\mathcal{J}$ on VOST validation set

TABLE VIII

ABLATION STUDIES ON COMPLEX DATASETS VOST AND MOSE FOR DIFFERENT COMPONENTS OF HIERARCHICAL CONTRASTIVE LEARNING LOSS, WHERE $\mathcal{L}_{ctr}^f$, $\mathcal{L}_{ctr}^p$, $\mathcal{L}_{ctr}^o$ DENOTE FRAME, PIXEL AND OBJECT-LEVEL LOSS RESPECTIVELY

ID	$\mathcal{L}_{ctr}^f$	$\mathcal{L}_{ctr}^p$	$\mathcal{L}_{ctr}^o$	VOST val		MOSE val		
				$\mathcal{J}$	$\mathcal{J}_{tr}$	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$
1				42.1	28.6	59.6	56.3	62.9
2		✓		44.6	30.4	61.8	58.4	65.2
3			✓	45.3	31.9	62.6	59.2	66.0
4	✓			46.9	33.2	64.4	60.8	68.0
5		✓	✓	48.0	35.3	66.5	63.2	69.8
6	✓	✓		48.4	35.9	67.1	63.6	70.6
7	✓		✓	52.3	40.1	72.3	68.1	76.5
8	✓	✓	✓	54.3	42.2	73.2	69.5	76.9

TABLE IX

ABLATION STUDIES ON COMPLEX DATASETS VOST AND MOSE THROUGH DIFFERENT SETTINGS OF OUR PROPOSED DYNAMIC MEMORY UPDATE (DMU) MODULES INCLUDING A SHORT-TERM MEMORY MODULATION (SMM) MODULE AND A LONG-TERM MEMORY AUGMENTATION (LMA) MODULE. THE SIMILARITY-WEIGHTED MODULATION (SWM), SHORT-TERM MEMORY PROPAGATION (SMP), VARIANCE PERCEPTION (VP) AND CONSISTENCY MAINTENANCE (CM) ARE THE CRUCIAL COMPONENTS OF SMM AND LMM MODULES. M_s AND M_l DENOTE SHORT-TERM AND LONG-TERM MEMORY.

ID	Setting	VOST val		MOSE val		
		$\mathcal{J} \uparrow$	$\mathcal{J}_{tr} \uparrow$	$\mathcal{J}\&\mathcal{F} \uparrow$	$\mathcal{J} \uparrow$	$\mathcal{F} \uparrow$
1	w/o M_s	46.4	33.1	62.9	58.1	67.7
2	w/o M_l	46.7	33.5	63.3	58.7	67.9
3	w/o DMU	47.2	34.4	63.5	58.9	68.1
4	w/o SMM	50.1	37.9	68.9	65.2	72.6
5	w/o SWM in SMM	51.8	39.1	70.7	67.8	73.6
6	w/o SMP in SMM	50.3	38.2	69.4	65.7	73.1
7	w/o LMA	51.3	38.8	70.1	67.0	73.2
8	w/o VP in LMA	52.9	40.7	71.8	68.1	75.5
9	w/o CM in LMA	52.4	39.1	70.6	67.3	73.9
10	Ours	54.3	42.2	73.2	69.5	76.9

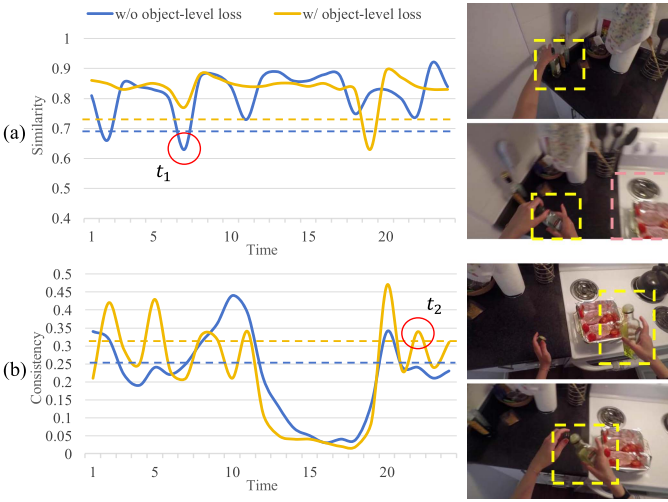


Fig. 8. Comparison of (a) vector similarity and (b) distribution consistency calculation process w/ and w/o object-level loss. The dotted lines are the pre-computed thresholds corresponding to the similarity and distribution respectively. The frames at t_1 , t_2 moments in red circle and their comparison frames for calculating similarity and consistency are shown on the right to illustrate the semantic and spatial changes.

which can adequately illustrate their effectiveness to amplify object changes at spatial and semantic aspects.

Effect of hierarchical contrastive learning: We conduct ablation studies on our proposed hierarchical contrastive learning including frame-level, pixel-level and object-level losses as shown in Table VIII. In setting 1, we only use the basic BCE loss to train our APC module which leads to significant accuracy decline as the consistency measurements are easily affected by background changes. Then we gradually add frame-level loss $\mathcal{L}_{ctr}^f$, pixel-level loss $\mathcal{L}_{ctr}^p$ and object-level loss $\mathcal{L}_{ctr}^o$ to construct the hierarchical contrastive learning loss. The evaluation results indicate that the frame-level, pixel-level and object-level losses all can improve the accuracy by effectively measuring the extent of spatial deformations and semantic transformations.

Furthermore, we visualize in Fig. 8 the similarity and consistency calculation processes before and after adding the object-level contrastive learning loss. In Fig. 8(a), when there are significant background changes at t_1 moment, for example, the

large distractor (in the pink box) at the bottom appears, the similarity results are slightly affected after adding object-level loss (shown in yellow line at t_1 moment), thus our proposed adaptive memory update mechanism can focus on the target objects as much as possible and be only sensitive to the object changes. In Fig. 8(b), when the target bottle (in the yellow box) is tipped over and mostly obscured by the hands at t_2 moment, the consistency before adding object-level loss is relatively high (the consistency score is close to 0 shown in blue line at t_2 moment) as most areas in the frame have not changed. In comparison, after adding object-level loss, the consistency becomes lower than the adaptively-decided threshold which indicates that the spatial changes of the target object is effectively captured by focusing on the object and the long-term memory will be updated at t_2 moment to store object variations. Therefore, with the introduction of object-level contrastive learning loss, our proposed hierarchical contrastive learning loss can enable the APC module to perceive and capture object semantic-spatial changes, and boost our CSNet to effectively deal with complex scenarios through the continuous attention on target objects throughout the videos.

Effect of dynamic memory update modules: We evaluate the effect of our proposed Dynamic Memory Update (DMU) modules including a Short-term Memory Modulation (SMM) module and a Long-term Memory Augmentation (LMA) module. As shown in Table IX, we first clarify the impact of short-term memory M_s and long-term memory M_l on segmentation performance in settings 1,2, and the results show that both short-term and long-term memory are effective to improve accuracy. Then we evaluate the overall impact of our proposed DMU modules by retaining the features directly in short-term and long-term memory instead of using DMU to enhance these features, and the results in setting 3 demonstrate that our proposed DMU modules are necessary to deal with complex scenarios. Finally, we evaluate the impact of SMM and LMA modules and their components, as shown in setting 4,5,6,7,8,9, these modules and

TABLE X
ABLATION STUDIES ON COMPLEX DATASETS VOST AND MOSE FOR THE
SAMPLING NUMBER K OF FRAMES FOR THE TRAINING AND INFERENCE
PROCESSES

K	VOST val		MOSE val		
	$\mathcal{J}$	$\mathcal{J}_{tr}$	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$
4	48.5	36.2	65.9	62.4	69.4
6	49.1	36.0	67.1	63.6	70.6
8	50.6	37.3	69.8	65.7	73.9
10	52.8	40.1	71.4	67.2	75.6
12	54.3	42.2	73.2	69.5	76.9

TABLE XI
RUNNING TIME (S) COST ANALYSIS OF EACH COMPONENT

	XMem	Cutie	Ours
Query encoder	0.861	0.851	0.853
Mask encoder	0.143	0.145	0.143
Pixel memory read	0.758	0.514	0.862
Object memory read	-	0.913	-
Decoding	2.749	0.725	0.727
Memory update judgment	-	-	0.641

their components are all crucial to dynamically update memory after adaptive clip division.

Effect of sampling frame number: We further clarify the effect of sampling number K of frames during the training and inference processes. As shown in Table X, we adopt 4,6,8,10,12 sampling number settings on the premise that the GPU memory can hold these frames. As the sampling number increases, the accuracy evaluated on the complex datasets VOST [29] and MOSE [30] gradually increases from 48.5% to 54.3% in terms of $\mathcal{J}$ on VOST validation set, thus we finally set the sampling number K to 12 during the training and inference processes of our APC module due to the GPU memory limitations.

Analysis of runtime cost: We conduct runtime cost analysis as shown in Table XI. Due to the lightweight architecture design of APC module to judge when to update memory and frame resize before inputting into APC, the runtime cost of our proposed memory update judgment is lower than the runtime cost of object memory read which is specifically designed in Cutie [19]. Besides, our DMU modules increase the runtime cost of pixel memory read is relatively small. Therefore, our method can achieve the best performance in a real-time speed of 31 FPS.

Analysis of failure cases: Furthermore, we observe that our method and existing SOTA methods struggle to the challenges including absent semantic and spatial overlap as shown in Fig. 9. For example, in case (a), only small parts of target objects (a car's wheel) are annotated in the first given mask, leading to significant lacking of the target objects. In case (b), similar objects (zebras) are mixed together, resulting in serious spatial overlap which is difficult for even humans to distinguish the real target object. Although our method can identify semantic and spatial changes of objects, the fundamental semantic deficiency and spatial confusion can still lead to terrible performance of existing methods.

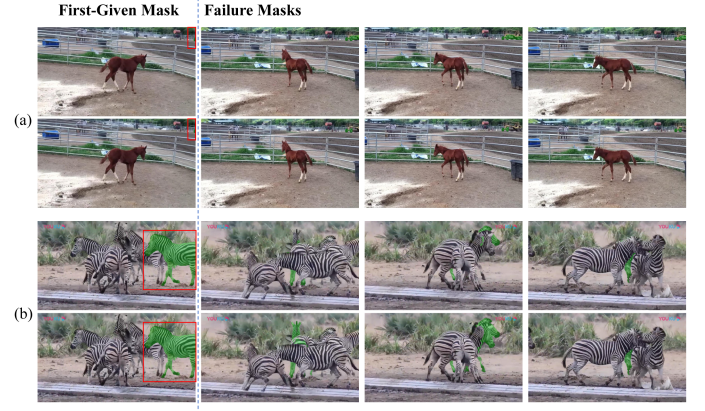


Fig. 9. Failure cases of our method (top) and Cutie (bottom) when facing severe absent semantic (a) and spatial overlap (b). We highlight the target object based on the first-given mask with red bounding box.

V. DISCUSSIONS

In previous memory-based methods [16], [18], [19], [20], [21], [23], [24], [67], [80], how to update memory has been studied intensively and each method contributes its own memory update approach, while the studies on when to update memory are relatively few as they usually adopt the simple fixed-interval memory update mechanism. However, according to the performance analysis on our proposed Change-Sensitive Network (CSNet), when dealing with intricate challenges in complex scenarios, it becomes particularly important to adaptively adjust the moments of memory updates based on object changes. Therefore, to fill in the gaps in existing Semi-VOS methods, we provide more in-depth analyses in this section on our proposed adaptive memory update mechanism which focuses on both when and how to update memory in our proposed CSNet.

Can the Adaptive Perception-Capture (APC) module work in different architectures? We further verify the versatility of our proposed Adaptive Perception-Capture (APC) module by plug-and-playing it into existing representative Semi-VOS methods [17], [18], [19]. As shown in Fig. 10, due to the independence of our APC module from encoding to decoding, we only need to adjust the moments of memory update according to the signal calculated by our APC module when integrated into existing methods.

After integrating our proposed APC module into existing representative methods including XMem [17], DeAOT [18] and Cutie [19] which have open-source codes, we conduct comprehensive experiments on common, complex and long-term datasets [26], [27], [29], [30], [34]. As shown in Table XII, after we replace the original fixed-interval memory update mechanism with our APC module in Cutie-R50 [19] method, there are 3.8%, 5.7% and 2.1% accuracy improvements in terms of $\mathcal{J}$, $\mathcal{J}_{tr}$, $\mathcal{J}\&\mathcal{F}$ on the validation sets of VOST and MOSE datasets respectively, and the speed is only reduced from 36 to 30 FPS which means our APC module has a minimal impact on the segmentation efficiency. As shown in Table XIII and Table XIV, when evaluated on common datasets, the introduction of our

TABLE XII

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS AFTER INTEGRATING OUR APC MODULE ON COMPLEX DATASETS VOST AND MOSE. WE RECORD THE INFERENCE SPEED (FPS) ON THE SAME A6000 GPU DEVICE AND * DENOTES THAT WE RETEST THE SPEED

Method	VOST val		VOST test		MOSE val			FPS
	$\mathcal{J}$	$\mathcal{J}_{tr}$	$\mathcal{J}$	$\mathcal{J}_{tr}$	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$	
CFBI [12]	45.0	32.0	43.9	32.1	45.7	41.2	50.1	5.9
CFBI+ [49]	46.0	32.6	46.7	31.6	47.8	42.9	52.7	5.6
STCN [16]	46.1	32.9	45.4	32.5	50.8	46.6	55.0	20.2
AOT-R50 [20]	48.7	36.4	49.9	37.1	57.2	53.1	61.3	18
RDE [23]	41.3	30.6	40.7	30.2	48.8	44.6	52.9	27
SWEM [24]	43.5	31.7	43.2	31.0	50.9	46.8	54.9	33
LLB [79]	42.4	31.9	41.9	30.2	52.0	48.6	55.5	8.3
AOT+ [29]	49.4	38.2	49.1	38.0	57.8	54.2	61.4	15*
XMem [17]	44.1	33.8	44.0	32.0	59.6	55.4	63.7	26*
XMem+APC	46.5	35.1	46.2	34.3	63.4	58.2	68.6	22
DeAOT-R50 [18]	50.4	36.1	50.3	36.2	64.1	59.5	68.7	27*
DeAOT-R50+APC	53.6	40.5	52.5	39.4	67.3	62.8	71.8	23
Cutie-R50 [19]	47.5	32.9	46.9	32.6	68.3	64.2	72.3	36*
Cutie-R50+APC	51.3	38.6	50.6	38.1	70.4	66.4	74.4	30

TABLE XIII

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS AFTER INTEGRATING OUR APC MODULE ON COMMON DATASETS INCLUDING DAVIS 2017 TEST AND YOUTUBE-VOS 2019 VAL

Method	DAVIS 2017 test			YouTube-VOS 2019 val			
	$\mathcal{J}\&\mathcal{F}\uparrow$	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	$\mathcal{G}\uparrow$	$\mathcal{J}_s\uparrow$	$\mathcal{F}_s\uparrow$	$\mathcal{J}_u\uparrow$
CFBI [12]	75.0	71.4	78.7	81.6	80.2	84.6	77.2
RMNet [77]	75.0	71.9	78.1	-	-	-	-
CFBI+ [49]	75.6	71.6	79.6	82.9	80.6	85.2	78.9
STCN [16]	76.1	72.7	79.6	82.7	81.1	85.4	78.2
AOT-R50 [20]	79.6	75.9	83.3	84.1	83.5	88.1	78.4
BATMAN [78]	82.2	78.4	86.1	85.0	84.5	89.3	79.0
RDE [23]	77.4	73.6	81.2	81.9	81.1	85.5	76.2
PMNet [92]	78.6	75.0	82.3	84.9	83.2	87.7	80.3
ISVOS [67]	82.8	79.3	86.2	86.1	85.2	89.7	80.7
XMem [17]	81.0	77.4	84.5	85.5	84.3	88.6	80.3
XMem+APC	81.8	78.2	85.4	85.9	85.1	89.5	80.4
DeAOT-R50 [18]	80.7	76.9	84.5	85.9	84.6	89.4	80.8
DeAOT-R50+APC	81.4	77.7	85.1	86.2	85.3	89.6	80.9
Cutie-R50 [19]	84.2	80.6	87.7	86.1	85.5	90.0	80.6
Cutie-R50+APC	84.5	81.1	87.9	86.6	86.1	90.4	81.2

TABLE XIV

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS AFTER INTEGRATING OUR APC MODULE ON COMMON DATASETS INCLUDING THE VALIDATION SETS OF DAVIS 2017 AND YOUTUBE-VOS 2018

Method	DAVIS 2017 val			YouTube-VOS 2018 val			
	$\mathcal{J}\&\mathcal{F}\uparrow$	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	$\mathcal{G}\uparrow$	$\mathcal{J}_s\uparrow$	$\mathcal{F}_s\uparrow$	$\mathcal{J}_u\uparrow$
CFBI [12]	81.9	79.3	84.5	81.8	81.9	86.3	75.6
RMNet [77]	83.5	81.0	86.0	81.5	82.1	85.7	75.7
CFBI+ [49]	82.9	80.1	85.7	82.8	81.8	86.6	77.1
STCN [16]	85.4	82.2	88.6	83.0	81.9	86.5	77.9
AOT-R50 [20]	84.9	82.3	87.5	84.1	83.7	88.5	78.1
BATMAN [78]	86.2	83.2	89.3	85.3	84.7	89.8	79.2
RDE [23]	84.2	80.8	87.5	-	-	-	-
PMNet [92]	87.1	83.5	90.7	85.1	84.1	88.8	79.7
ISVOS [67]	87.1	83.7	90.5	86.3	85.5	90.2	80.5
XMem [17]	86.2	82.9	89.5	85.7	84.6	89.3	80.2
XMem+APC	86.4	83.1	89.7	85.9	84.8	89.6	80.3
DeAOT-R50 [18]	85.2	82.2	88.2	86.0	84.9	89.9	80.4
DeAOT-R50+APC	85.7	82.8	88.6	86.1	85.1	90.0	80.5
Cutie-R50 [19]	88.8	85.4	92.3	86.1	85.8	90.5	80.0
Cutie-R50+APC	89.1	85.7	92.5	86.2	85.9	90.6	80.1

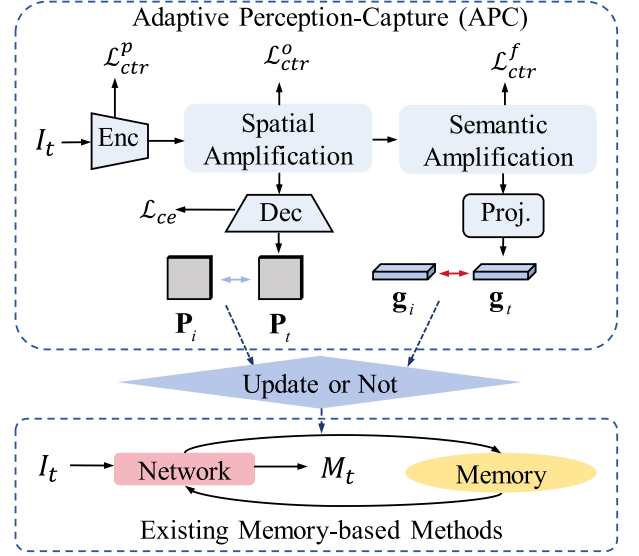


Fig. 10. Illustration of how our proposed Adaptive Perception-Capture (APC) module plug and plays into existing matching-based methods with memory network. I_t , M_t denote the input frame and generated mask, P_i , P_t denote the generated probability distribution of the initial and current frame in the i -th clip, g_i , g_t denote the generated global vector of the initial and current frame in the i -th clip. The only adjustment to the existing memory-based Semi-VOS methods is to replace original fixed-interval memory update mechanism with our adaptive memory update mechanism by updating memory based on the update signals of our proposed APC module.

TABLE XV

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS AFTER INTEGRATING OUR APC MODULE ON LONG-VIDEO DATASET LVOS. MB DENOTES THE KIND OF MEMORY BANK. F+P, A, C AND LT DENOTE FIRST AND PREVIOUS FRAME, ALL FRAMES, COMPRESSED MEMORY AND LONG-TERM MEMORY RESPECTIVELY. MEM(G): MAXIMUM GPU MEMORY USAGE.

Method	MB	LVOS val			LVOS test			Mem(G)
		$\mathcal{J}\&\mathcal{F}\uparrow$	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	$\mathcal{J}\&\mathcal{F}\uparrow$	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	
CFBI [12]	F+P	51.5	46.2	56.7	44.8	40.2	49.4	3.82
AFB-URR [32]	C	36.2	33.1	39.3	40.8	37.5	44.1	2.89
STCN [16]	A	48.9	43.9	54.0	48.3	44.0	52.5	0.92
AOT-L [20]	A	60.9	55.1	66.8	54.7	49.2	60.2	1.32
RDE [23]	C	53.7	48.3	59.2	50.2	45.7	54.6	1.0
XMem [17]	LT	52.9	48.1	57.7	50.9	46.5	55.3	3.34
XMem+APC	LT	53.6	48.9	58.3	51.4	47.1	55.7	3.67
Cutie-R50 [19]	LT	60.1	55.9	64.2	56.2	51.8	60.5	2.36
Cutie-R50+APC	LT	60.5	56.4	64.6	56.3	52.0	60.6	2.52
DDMemory [34]	C	61.9	56.3	67.4	55.7	50.3	61.2	0.88
DDMemory+APC	C	62.2	56.6	67.8	55.9	50.5	61.3	0.91

APC module can further improve the accuracy due to proactively updating memory. As shown in Table XV, our APC module can consistently improve the segmentation accuracy without too much extra memory usage when dealing with long-term videos. Therefore, through incorporating our APC module into existing memory-based Semi-VOS methods with different architectures, the versatility of our APC module has been sufficiently verified which can significantly improve accuracy without sacrificing too much efficiency and memory usage when dealing with different kinds of datasets.

How does our adaptive memory update mechanism perform compared to other memory update mechanisms? To clearly

TABLE XVI
QUANTITATIVE COMPARISONS OF DIFFERENT MEMORY UPDATE MECHANISMS
BASED ON EXISTING REPRESENTATIVE MEMORY-BASED METHODS ON THE
COMPLEX DATASETS VOST AND MOSE

Method	VOST val		MOSE val		
	$\mathcal{J}$	$\mathcal{J}_{tr}$	$\mathcal{J} \& \mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$
STM [15]	41.2	27.5	39.6	36.3	42.9
STM+AFB [32]	40.9	26.6	39.4	36.0	42.8
STM+PAM [31]	35.9	20.1	35.4	31.7	39.1
STM+QAM [70]	43.3	29.1	40.9	36.5	45.3
STM+Ours	44.1	30.9	42.7	38.3	47.1
STCN [16]	46.1	32.9	50.8	46.6	55.0
STCN+RDE [23]	41.3	30.6	48.8	44.6	52.9
STCN+SWEM [24]	43.5	31.7	50.9	46.8	54.9
STCN+CorrLearn [59]	46.6	33.3	52.0	48.2	55.8
STCN+Ours	47.3	34.1	53.1	49.3	56.9
XMem [17]	44.1	33.8	59.6	55.4	63.7
XMem+CorrLearn [59]	44.8	34.3	61.1	56.8	65.4
XMem+Ours	46.5	35.1	63.4	58.2	68.6
AOT-R50 [20]	48.7	36.4	57.2	53.1	61.3
AOT-R50+RMem [21]	50.5	39.8	58.6	54.5	62.7
AOT-R50+Ours	51.6	40.1	59.8	55.6	64.0
DeAOT-R50 [18]	50.4	36.1	64.1	59.5	68.7
DeAOT-R50+RMem [21]	51.8	40.4	66.0	61.1	70.9
DeAOT-R50+Ours	53.6	40.5	67.3	62.8	71.8

illustrate the advantages of our proposed adaptive memory update mechanism, we further conduct extensive experiments to compare our proposed mechanism with existing memory update mechanisms [21], [23], [24], [31], [32], [59], [70] which have open-source codes. Table XVI presents the quantitative comparisons of different memory update mechanisms when evaluated on the complex datasets [29], [30] based on different representative Semi-VOS methods [15], [16], [17], [18]. For each baseline method, we integrate existing representative memory update mechanisms and our adaptive memory update mechanism respectively. For example, based on the classical memory-based method STM [15], our mechanism can improve more accuracy compared to the QAM mechanism [70] (2.1% vs 2.9% improvements in terms of $\mathcal{J}$ on VOST validation set). Moreover, based on a stronger baseline method DeAOT [18], our mechanism can boost the performance to 53.6% and 67.3% when evaluated on VOST and MOSE validation set, which achieves significant improvements compared to existing memory update mechanism RMem [21]. Through comprehensively comparing the effects of different memory update mechanisms on segmentation performance, the advantages of our proposed adaptive memory update mechanism are further highlighted when dealing with intricate challenges in complex scenarios.

Can the adaptive memory update mechanism be extended to other tasks? Due to the temporal continuity of videos, similar to storing historical features as memory to boost matching in the VOS task, it is natural to exploit historical informations to better process the current frame in video-related task [93], [94], [95]. Therefore, we further extend our proposed adaptive memory update mechanism to the visual tracking task to adequately illustrate that our proposed mechanism is also adaptable to other video-related tasks. Specifically, we replace

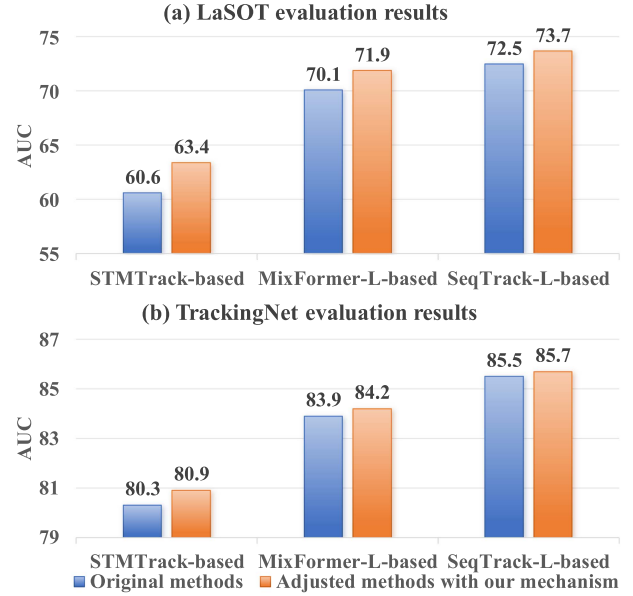


Fig. 11. Quantitative comparisons between existing visual tracking methods and adjusted methods by replacing original template online update strategies with our proposed adaptive memory update mechanisms when evaluated on (a) LaSOT and (b) TrackingNet dataset.

the template online update modules in existing visual tracking methods [96], [97], [98] with our proposed APC module, and evaluate these adjusted methods on LaSOT [99] and TrackingNet [100] datasets which consists of long-term and complex videos. Meanwhile, we replace the training losses for the original template online update modules with our proposed hierarchical contrastive learning losses. As shown in Fig. 11, we intuitively present the quantitative comparisons between original methods and adjusted methods with our mechanism in terms of AUC after re-training adjusted methods following the same training setting of the original visual tracking methods. The results clearly demonstrate that our proposed mechanism can be applied to existing visual tracking methods and also effectively improve the tracking performance compared to their original template online update modules.

VI. LIMITATIONS AND FUTURE WORKS

Although our proposed Change-Sensitive Network (CSNet) effectively improves the performance when evaluated on common, complex and long-term datasets, there are still significant gaps between the segmentation accuracy of common and complex, long-term datasets (87.2% on YouTube-VOS 2019 val [27], 54.3% on VOST val [29], 58.9% on LVOS test [34] shown in Tables I, III and IV). One possible solution to bridge these gaps is to adopt stronger backbone such as Swin [101], ConvViT [91]. However, these stronger backbones usually lead to unsatisfactory efficiency, for example, STMA [80] using ConvViT-B segments objects at a relatively low speed of 14 FPS as shown in Table I. Meanwhile, resource-constrained deployment environments require lightweight and energy-efficiency model variants, which motivates us to design effective architecture

for better utilizing input stream to reduce energy consumption following [102], [103], [104]. Therefore, we will further dig into how to achieve higher accuracy when segmenting objects in complex and long-term datasets under the premise of real-time segmentation.

VII. CONCLUSION

In this paper, we propose a novel Change-Sensitive Network (CSNet) to learn when and how to update memory for semi-supervised video object segmentation. To perceive and capture object spatial-semantic changes, we design an Adaptive Perception-Capture Module with hierarchical contrastive learning at frame, object and pixel levels to discern the extent of spatial deformations and semantic transformations of target objects, thus we can master temporal patterns of varying videos and divide videos into clips. Based on the effective division of clips, we innovatively construct dynamic memory update modules including a Short-term Memory Modulation module to extract object prototypes within clips and a Long-term Memory Augmentation to enhance object variances across clips. In this way, our proposed CSNet can achieve significant improvements when dealing with intricate challenges in emerging complex scenarios. Furthermore, we sufficiently discuss the adaptive memory update mechanism in our proposed CSNet and illustrate that our mechanism can be applied to different Semi-VOS methods and other video-related tasks. Extensive and comprehensive evaluation results demonstrate the effectiveness of our proposed CSNet and adaptive memory update mechanism. For the video-related tasks, our proposed mechanism can act as a brand-new, flexible and effective way to deal with historical informations of videos and provide abundant exploration spaces for future studies.

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